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Agents and Pervasive Computing Group

Autonomic : a Booster for Pervasive

Franco Zambonelli
Università di Modena e Reggio Emilia
franco.zambonelli@unimore.it



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Autonomic Communication: a Booster for Next Generation Pervasive Ecologies

Franco Zambonelli
Università di Modena e Reggio Emilia
franco.zambonelli@unimore.it



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Where are We Now?

- PerCom
  - Just having a look at most papers around…
  - Technology
  - Inter-operability and spontaneous networking
  - New paradigms of human-environment interaction
- Autonomic Computing
  - Advanced autonomous/intelligent management of clusters
  - Knowledge-based and Agent-based approaches
- In all cases
  - Rather static application-specific systems
  - Working in isolation and small-medium size
  - Humans may enforce control



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Where are We Headed?

- These areas are paving the way towards
  - The dense deployment of computational and sensing devices
  - Widely used and usable, and for the use of everybody
  - No one will have control
- In such a scenario
  - Scale grows to dramatic numbers
  - No system will be isolated, and multiple “specimens” of applications, services, data, will live in such scenarios
  - Like in an ecological system
- This will limit the effectiveness of
  - Close feedback loops for optimization, planning, etc.
  - As studied in mainstream autonomic computing
  - And assuming the possibility of direct control and intervention



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Autonomic Communication

- Focuses on
  - Context-aware and flexible approaches to networking
  - Assumes no way of controlling the network at a fine-grained level
  - Humans are really out of the loop
  - (Nature-inspired) Self-organization to enforce distributed functionality
  - (Socially-inspired) Self-preservation to achieve robustness
- Cfr. The "Situated and Autonomic Communication" EU Initiative
  - The CASCADAS Project www.cascadas-project.org
- These will be really needed for emerging pervasive ecologies!
  - In addition to inter-operability, technology, etc.
- But still may not be enough…



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Autonomic Challenges for Pervasive Ecologies

- Developing and Deploying Application and Systems
  - Conceiving them as non-autochthonic specimen
  - Accounting for their impact on the surrounding ecosystem
  - …and viceversa
  - No system will be an island
- Exploiting Self-organization in an Engineered Way
  - Not simply mimicking nature and reverse engineering
  - But direct design of self-organizing systems
- Defending from Self-organizing (emergent) Behaviors
  - Decentralized and Integrated approaches to control